

A close-up photograph of a man's face, partially obscured by a semi-transparent blue overlay. He is sneezing, with his eyes closed and his hands pressed against his nose and mouth. The background is white with abstract yellow and blue geometric shapes on the right side.

Rinitis Alergi

Pendahuluan

- ▶ Rinitis alergi (RA) □ **penyakit inflamasi di mukosa hidung**, diperantarai imunoglobulin E (IgE) akibat **pajanan alergen yang dihirup** oleh individu yang sensitif.¹
- ▶ Keadaan inflamasi kronik pada mukosa hidung □ **inflamasi saluran nafas atas yang mempunyai epitel yang sama, lokasi yang berdekatan** □ menjadi **komorbiditas rinitis alergi**.²
- ▶ Keluhan yang paling sering pada orang dewasa dan anak-anak adalah hidung tersumbat.

1.Hellings PW. What is allergic rhinitis. EAACI, 2015, pp.2-4.

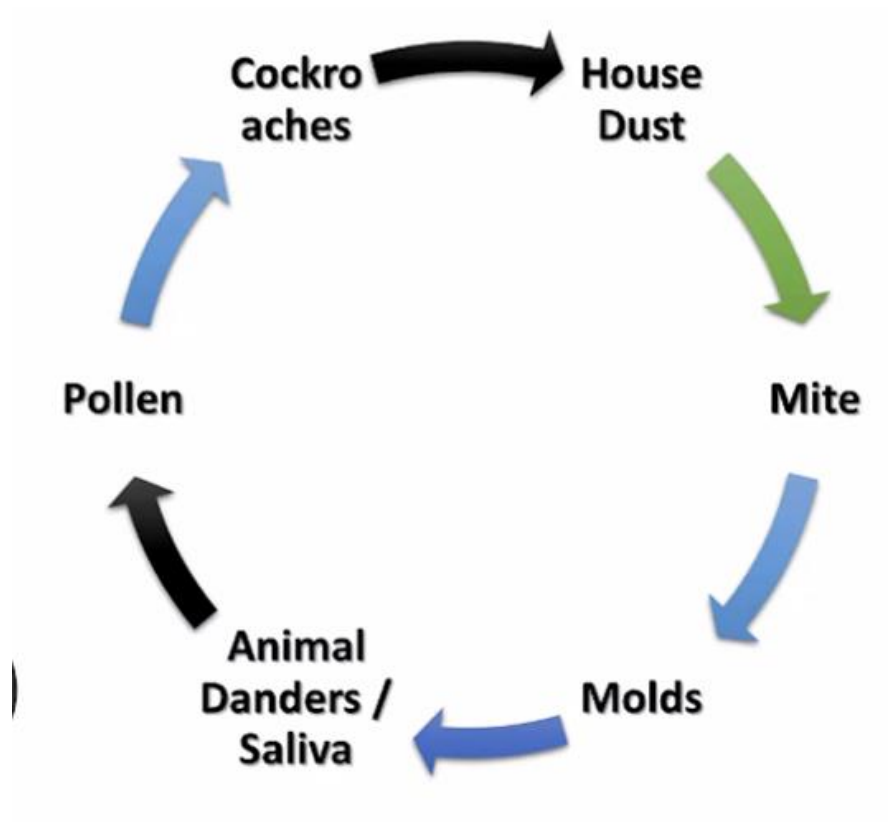
2.Mailasari A. Penegakan Diagnosis Penyakit THT yang berkaitan dengan Rinitis Alergi. 2022

Epidemiologi

- ▶ Di Indonesia prevalensi rinitis alergi meningkat 1,5-12,4%^{↑↑} dari seluruh populasi, walau tidak menimbulkan kematian, dampaknya dapat menurunkan kualitas hidup manusia.¹
- ▶ Rhinitis alergi diderita 10-30% orang dewasa dan lebih dari 40% pada anak di Amerika.²

1. Prepageran N. Managing allergic rhinitis and chronic rhinosinusitis in developing countries. EAACI, 2015, pp.400-403
2. Pawankar R, et al. WAO White Book on Allergy: Update 2013 Executive Summary. 2013

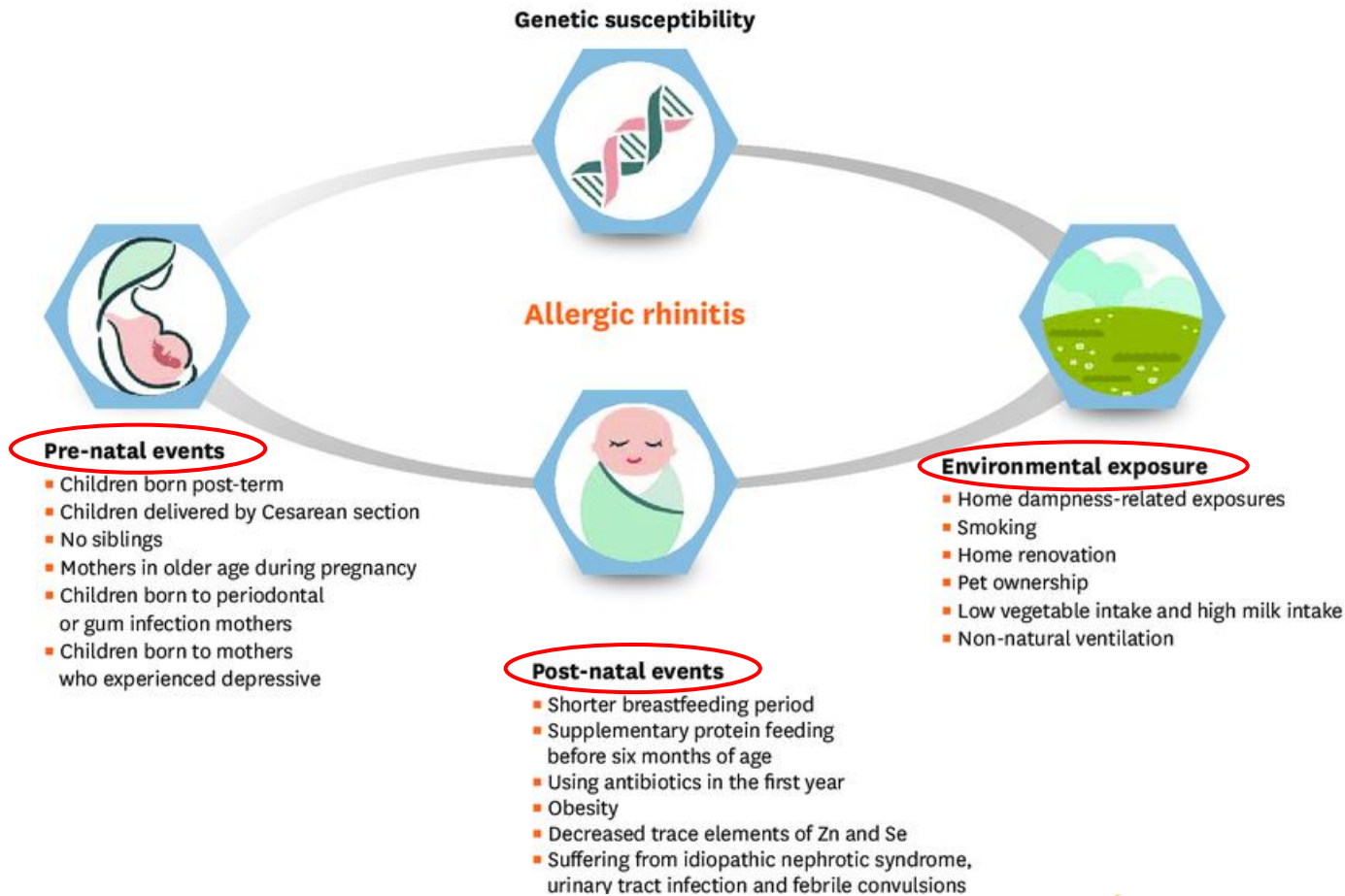
Etiologi: Triggering Rinitis Alergi



Faktor Resiko

Changing Allergic Rhinitis in China

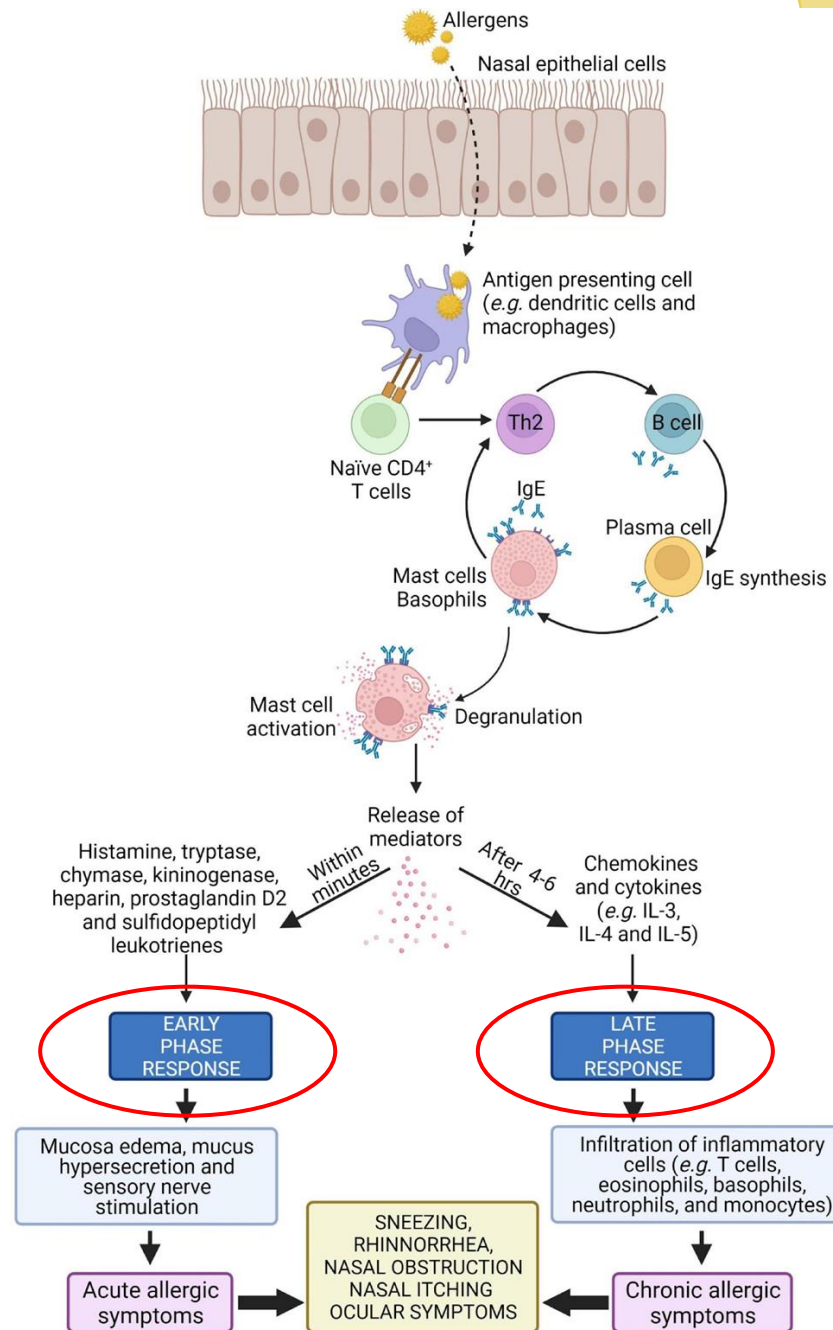
Allergy, Asthma & Immunology Research **AAIR**



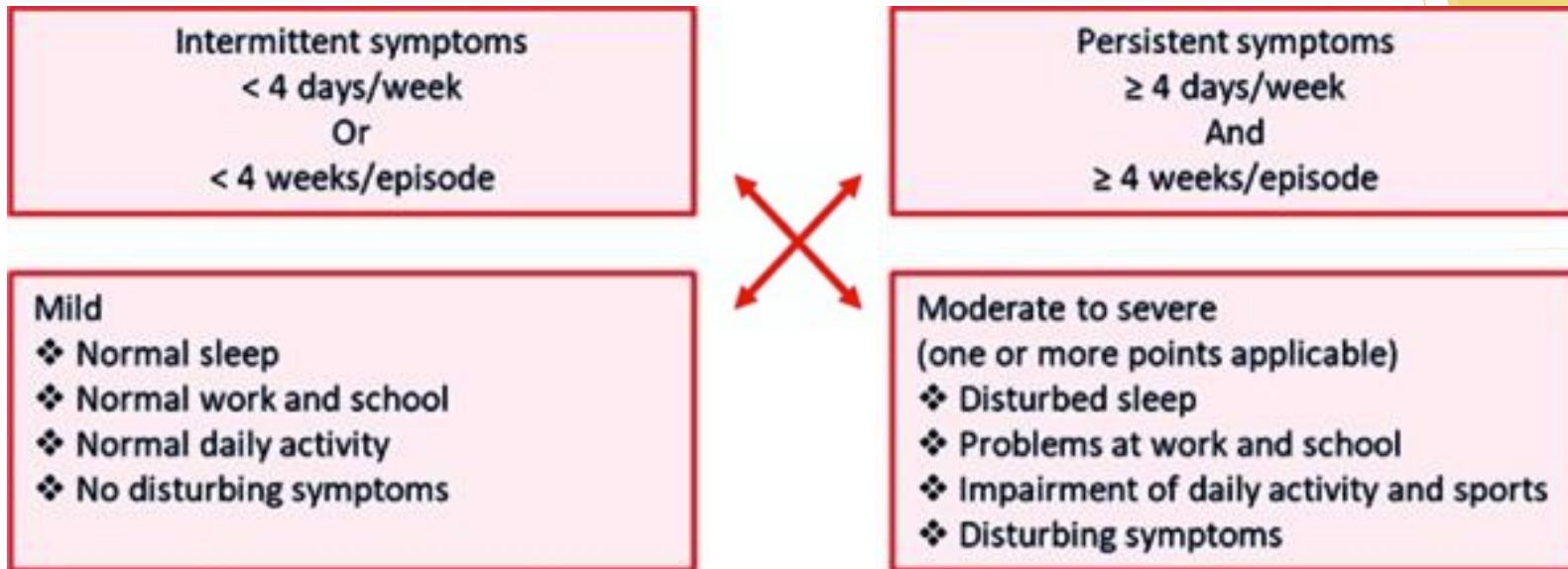
Patofisiologi



Patofisiologi



Klasifikasi

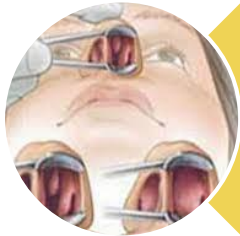


Bousquet J, et al. Allergic Rhinitis and its Impact on Asthma (ARIA) 2008 update
EAACI 2008; 63.

Diagnosis



Anamnesis



Pemeriksaan
Fisik



Pemeriksaan
Penunjang

Table 2**Components of the history and physical examination that may aid in identification of etiologies of nasal obstruction**

History	Physical Examination
Nature of nasal obstruction	Complete head and neck examination
Unilateral vs bilateral	Evidence of atopy (conjunctivitis, allergic shiners, supratip crease)
Perennial vs seasonal	Evidence of malignancy (lymphadenopathy, cranial nerve deficit)
Timing throughout day	
Relation to environmental exposures	
Duration	
Associated sinonasal symptoms	Nasal examination
Hyposmia/anosmia	External examination (nasal deformity, dynamic collapse of nasal sidewall)
Nasal discharge (purulent vs other)	Anterior rhinoscopy (inferior turbinates, septum, internal nasal valve, evidence of inflammation)
Facial pain/pressure	Modified Cottle maneuver
Other atopic symptoms	Nasal endoscopy
Sneezing	Septal deviation
Itchy, watery eyes	Turbinate enlargement
Eczema	Mass or polyp
Asthma	Mucosal edema
Food allergy	Discharge
Known environmental triggers	Scarring
History of sinonasal surgery or trauma	Crusting
Medications	Dynamic collapse of internal valve
Topical	Foreign body
Systemic	Adenoid enlargement
Social history	
Tobacco use	
Illicit intranasal drug use	
Work/occupational exposures	
Pregnancy	

Gejala yang berhubungan dg Rinitis Alergi

- 2 atau lebih dari keluhan yang berlangsung >1 jam dalam sehari:
 - Rinorea watery / encer
 - Bersin, dan beruntun / paroksismal
 - Kongesti hidung
 - Nasal pruritus $\pm$ konjungtivitis

Gejala yang TIDAK berhubungan dg RA

- Keluhan unilateral
- Kongesti nasal tanpa keluhan lain
- Mukopurulen rinorea
- Post nasal drip dengan secret kental dan/ tanpa rinorea anterior
- Nyeri
- Epistaksis berulang
- anosmia

Visual Analog Scale (VAS) Rinitis Alergi

Patient's page

0 1 2 3 4 5 6 7 8 9 10

Not troublesome at all

How are you feeling today?

Think about how troublesome your symptoms have been for the last 24 h

Very troublesome

Nasal symptoms:

- congestion
- itching
- secretion
- sneezing

Ocular symptoms:

- redness
- watery eyes
- itching

Doctor's page

10 9 8 7 6 5 4 3 2 1 0

Not troublesome at all

Classification of allergic rhinitis control

Very troublesome

>5 uncontrolled

<2 to ≤5 partially controlled

<2 controlled

Yang menilai adl pasien
Dari Skala 0-10, bagaimana
gejala dan tanda RA
mengganggu hari ini

1. 0 = tidak mengganggu
2. 10 = sangat mengganggu

Nilai VAS dapat
diklasifikasikan secara
retrospektif, dengan membuat
kelompok nilai

- VAS ≥ 5 = RA tidak terkontrol
- VAS 2- <5 = RA terkontrol Sebagian
- VAS <2 = RA terkontrol baik

Visual Analog Scale (VAS) Rinitis Alergi

The image displays two parts of a Visual Analog Scale (VAS) for Allergic Rhinitis control.

Patient's page: This section features a horizontal scale from 0 to 10. At 0, it says "Not troublesome at all" and at 10, it says "Very troublesome". The central text asks "How are you feeling today?" and instructs the patient to "Think about how troublesome your symptoms have been for the last 24 h". To the right, it lists symptoms: "Nasal symptoms: congestion, itching, secretion, sneezing" and "Ocular symptoms: redness, watery eyes, itching".

Doctor's page: This section features a horizontal scale from 10 to 0. At 10, it says "Not troublesome at all" and at 0, it says "Very troublesome". The central text asks "Classification of allergic rhinitis control". Below the scale, there are three colored boxes indicating control levels: a red box for ">5 uncontrolled", a yellow box for "<2 to ≤5 partially controlled", and a green box for "<2 controlled".

Gejala:

1. Hidung buntu
2. Hidung Gatal
3. Rinorea
4. Bersin

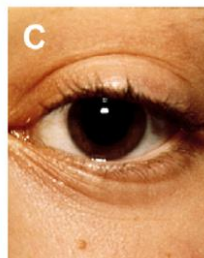
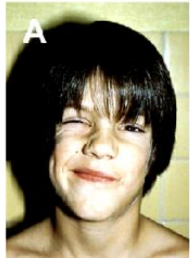
Keluhan mata

1. Kemerahan
2. Mata berair
3. Mata Gatal

Sybilski A.J. Visual analogue scale. A simple tool for daily treatment monitoring in allergic rhinitis. PEDIATR MED RODZ Vol. 14 No. 3, p. 277-281 . 2018

Diagnosis

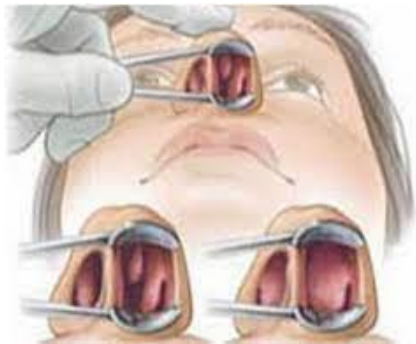
► Pemeriksaan Fisik



- A) *Facial grimacing or twitching* (gatal pada hidung);
- B) *allergic shiner*;
- C) *Dennie-morgan lines*
- D) *Allergic salute*;
- E) *Nasal creasing related to allergic salute*;
- F) *Allergic facies*;
- G) *Mukosa hidung pucat, berair, edema dilihat dari nasoendoskopi*

Diagnosis

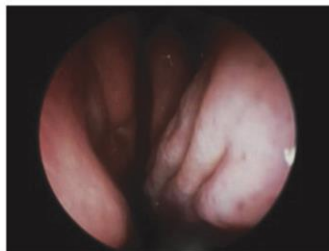
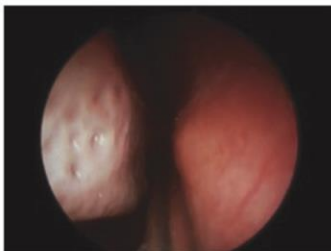
► Pemeriksaan Fisik



Stage 1

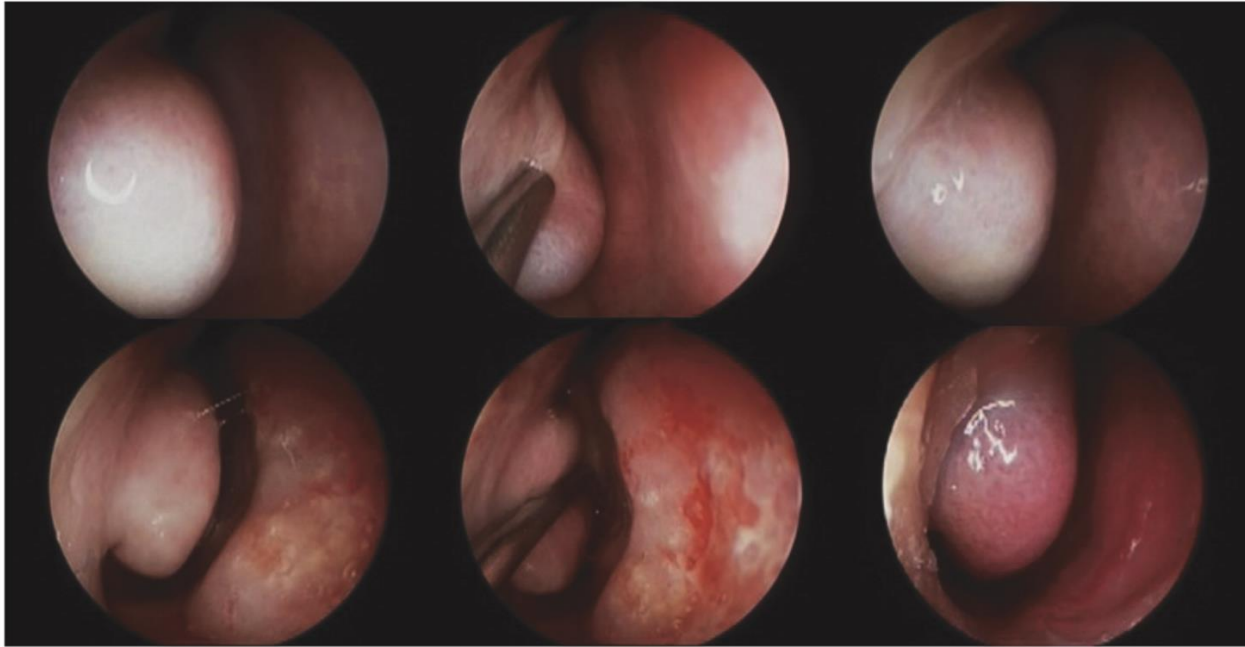
Stage 2

Stage 3



- Rinoskopi anterior dan endoskopi
- Harus dievaluasi sebelum dan sesudah pemberian topikal dekongestan
- Untuk menilai kontribusi edema mukosa reversibel yang menyebabkan hidung buntu

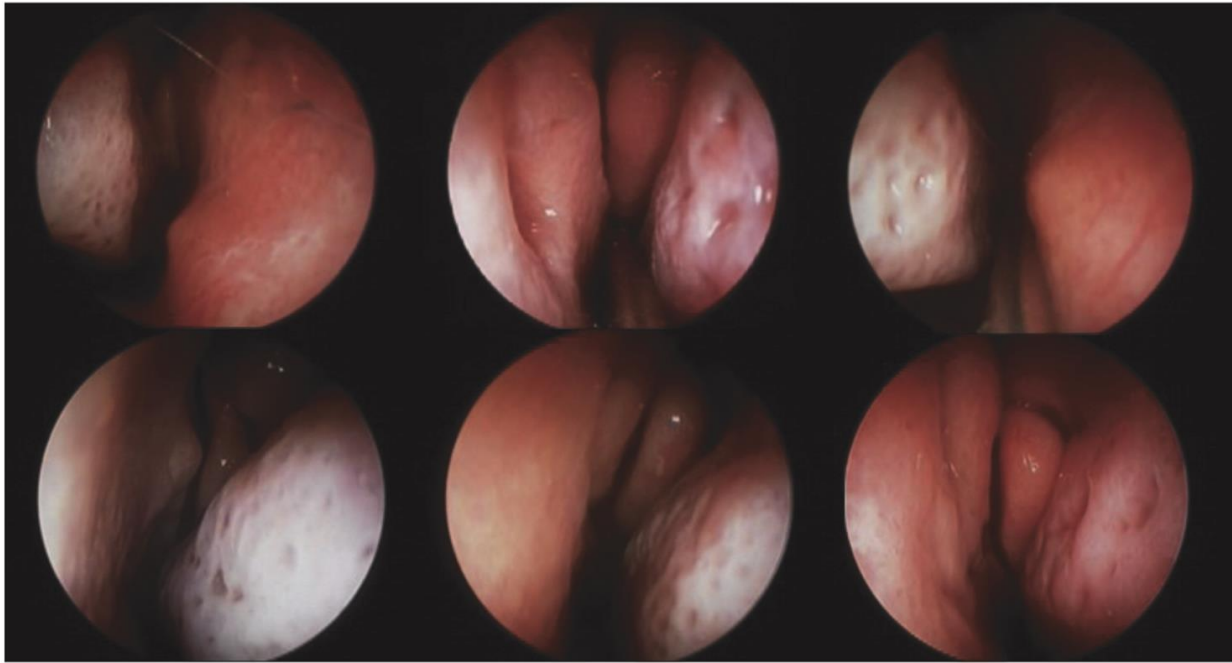
Fig. 2. Three Stage of Allergic Rhinitis – DANYOUNG Classification by Nasoendoscopic Video System



Hipertrofi

Fig. 3. Stage 1. Hypertrophy State of Allergic Rhinitis – DANYOUNG Classification by Nasoendoscopic Video System

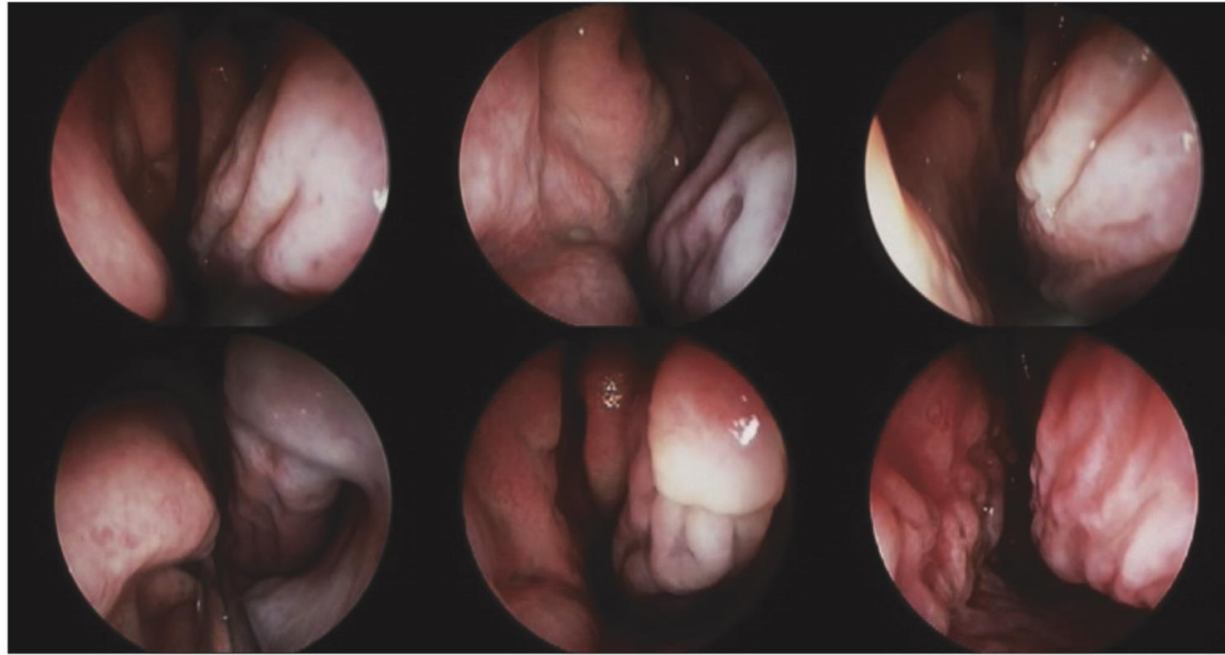
Chang, Sun-Ho. (2015). Newly Allergic rhinitis classification by Innovative Endoscopic Diagnostic Method - Danyoung Classification. Head Mirror Journal. Vol.1. 23-30.



Lesung Pipi

Fig. 4. Stage 2. Dimple State of Allergic Rhinitis – DANYOUNG Classification by Naso-endoscopic Video System

Chang, Sun-Ho. (2015). Newly Allergic rhinitis classification by Innovative Endoscopic Diagnostic Method - Danyoung Classification. Head Mirror Journal. Vol.1. 23-30.



Kerutan

Fig. 5. Stage 3. Wrinkle State of Allergic Rhinitis – DANYOUNG Classification by Nasoendoscopic Video System

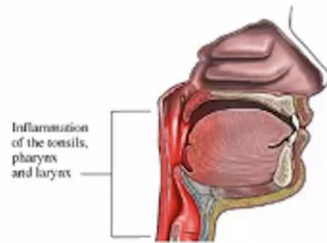
Chang, Sun-Ho. (2015). Newly Allergic rhinitis classification by Innovative Endoscopic Diagnostic Method - Danyoung Classification. Head Mirror Journal. Vol.1. 23-30.

Pemeriksaan Penunjang



- *Skin Prick Test* (SPT) dan serum spesifik IgE.
- SPT merupakan *gold standart* untuk mendiagnosis rinitis alergi
- Tes IgE spesifik dengan RAST sering digunakan ☐ penderita dengan penyakit kulit atau yang tidak dapat berhenti obat

The Correlation Between Allergic Rhinitis & Respiratory Tract



Upper
Respiratory
Tract Infection

Otitis media
with effusion



Nasal
Polyp

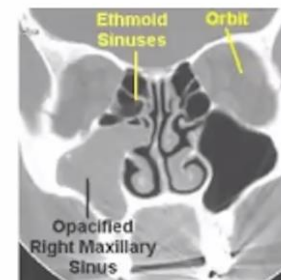


**Allergic
Rhinitis**

Asthma

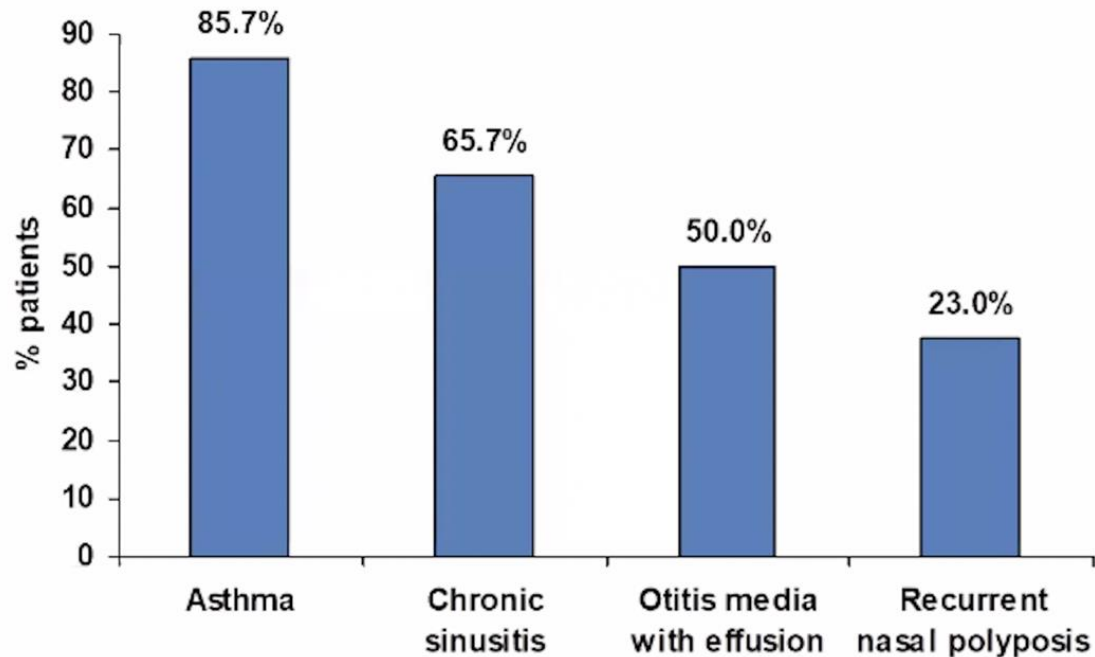


Rhino
sinusitis



Komorbid Rinitis Alergi

Patients (%) with selected conditions + co-morbid



Allergic rhinitis is a common co-morbidity in patients with other upper respiratory tract conditions

RA dan Asma

- ▶ Hubungan antara asma dan rhinitis alergi karena peradangan mukosa saluran nafas yang sama pada kedua penyakit tersebut
- ▶ Paparan alergen yang saluran nafas merupakan factor resiko tersering pada asma
- ▶ Penelitian menunjukkan terapi RA dapat mengurangi tingkat keparahan dari asma

RA dan Rhinosinusitis

- ▶ Rinitis alergi membuat hipertrofi konka secara kronik □ menutup osteometal kompleks
- ▶ Keadaan akut □ edema konka, hipersekresi, infiltrasi eosinophil □ siliostasis □ sekresi menumpuk □ berkelanjutan infeksi
- ▶ Menurut studi Skandinavia, alergi adalah faktor yang mendasari 40% hingga 67% pasien sinusitis kronis

RA dan Otitis Media Efusi (OME)

- ▶ Kondisi pada telinga tengah, disertai disfungsi tuba eustachius dan tekanan negatif telinga tengah akibat pelepasan mediator dan sitokin oleh sel mast dan sel radang lain yang berlangsung lama dan terus menerus
- ▶ Terkait dengan penurunan pendengran t.u pada anak
- ▶ Prevalensi RA pada anak-anak dengan OME berkisar dari 35% hingga 50%
- ▶ Penelitian menemukan □ peningkatan kadar sitokin Th-2, IL4, IL6, dan TNF α dalam efusi telinga tengah pada OME dengan rinitis alergi

RA dan ISPA

- ▶ Porro et al, mengidentifikasi atopi sebagai pemicu infeksi saluran pernapasan atas pada masa anak-anak
- ▶ Paparan alergen dapat menginduksi ekspresi molekul adhesi antar sel, yang merupakan reseptor untuk 90% rhinovirus pada manusia

RA dan Polip

- ▶ Tingkat kekambuhan dalam sebuah penelitian terhadap 140 pasien adalah 36% pada individu alergi versus hanya 18% pada individu yang tidak alergi.
- ▶ Hubungan alergi dengan polip hidung karena adanya eosinofil, sitokin, dan sel inflamasi yang khas pada alergi
- ▶ Penelitian mengevaluasi 211 pasien dengan polip hidung: 71% menderita asma, 76% menjalani tes kulit positif

Tatalaksana

- ▶ Menghindari alergen
- ▶ Terapi medikamentosa
- ▶ Immunoterapi

Terapi bersifat mengontrol alergi dan mengembalikan fungsi pernafasan atas

Elimination and avoidance of antigens.

<Elimination of house dust mites>

1. For indoor cleaning, use an exhaust circulation-type cleaner. Clean room for 20 s/m² twice a week.
2. Avoid using textile sofa, carpet, and tatami wherever possible.
3. Put antimite covers over mattresses, beds, and pillows.
4. Keep humidity at 50% and room temperature at 20–25 °C.

<Avoidance of cedar pollen>

1. Collect pollen information.
2. Stay at home during a heavy pollen dispersal period.
3. Shut windows and doors during a heavy pollen dispersal period.
4. When going out during a heavy pollen dispersal period, wear a mask and glasses.
5. When going out, avoid wearing woolen coats.
6. When going home, shake dust off from suit and hair before entering. Wash the face, gargle, and blow your nose.
7. Clean rooms frequently.

<Reduce pet (especially cat) antigens>

1. Stop keeping pets if possible.
2. Keeps pets outdoors and keep them away from bedroom.
3. Clean pets and their environments.
4. Change carpet to flooring.
5. Improve ventilation, and clean rooms.

Adapted from Practical Guideline for the Management of Allergic Rhinitis in Japan 2009, digest version.

Avoiding Allergic Triggers



Wash bedding
regularly



Dust frequently



Rinse nasal passages



Wash hands after
playing with pets



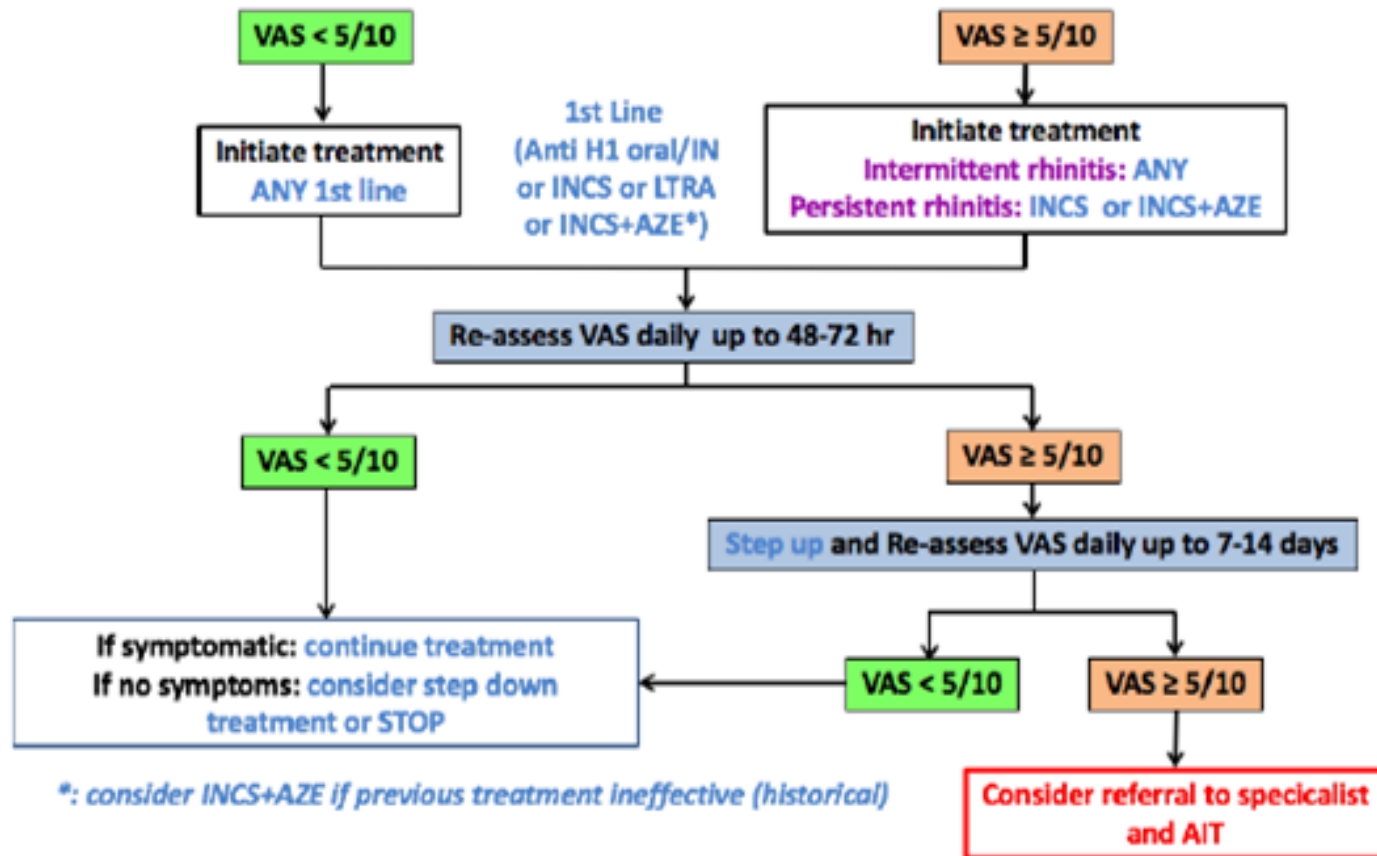
Shut windows during
pollen season



Avoid smoke and
fragrances

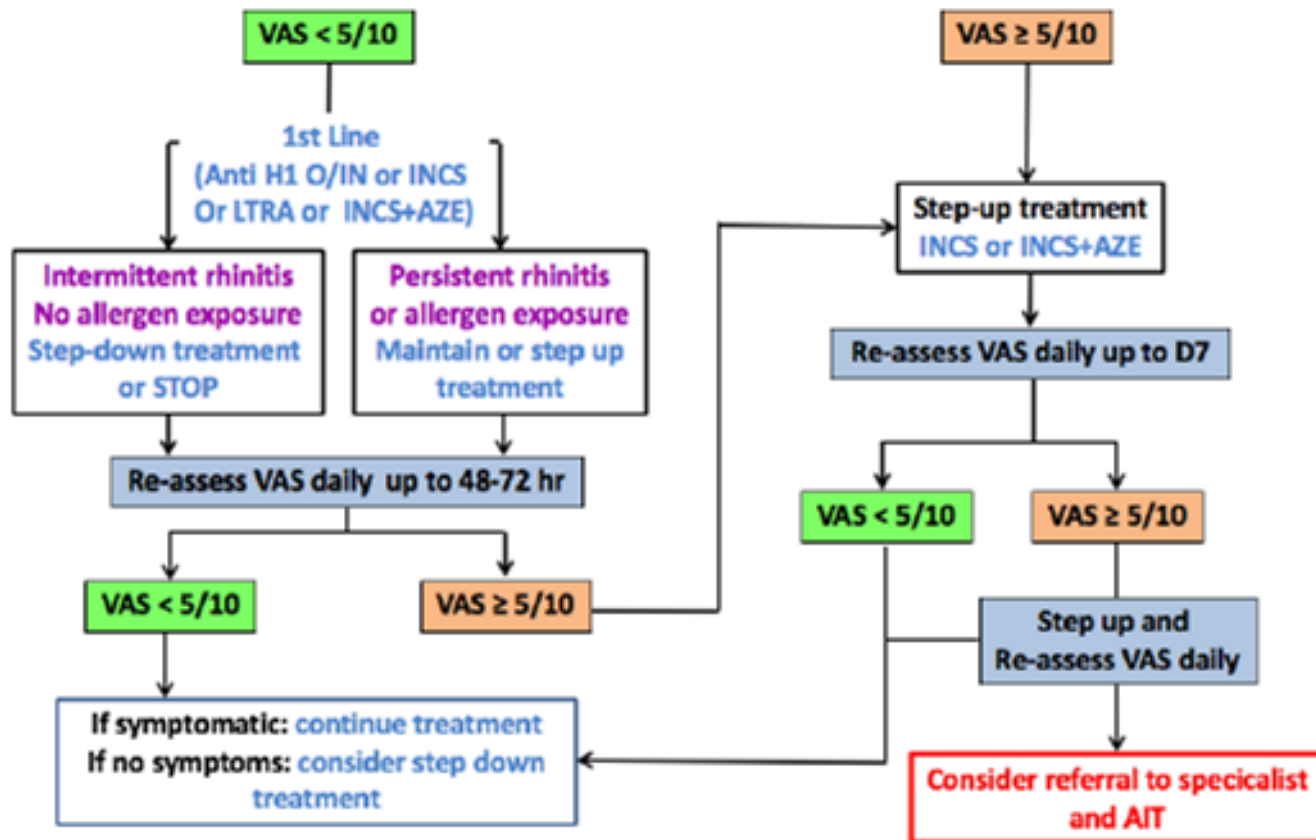
Tatalaksana

Assessment of control in untreated symptomatic patient



Tatalaksana

Assessment of control in treated symptomatic patient



Treatment Initiation

Patients	Consensus	Note
IAR; VAS ≥ 5	T1, T2, or T3	T2 or T3 if congestion
PER; VAS ≥ 5	T2 or T3	
IAR or PER VAS < 5	T1, T2, or T3	T2 or T3 if congestion

Classification of treatment used in patients with allergic rhinitis
T1 Nonsedating H1-antihistamine(oral, intranasal, and ocular), leukotriene receptor antagonist, or cromones (intranasal and ocular) T2 INCSs T3 INCSs + intranasal azelastine T4 Oral corticosteroid as a short course and an add-on treatment T5 Consider referral to a specialist and allergen immunotherapy

Consensus opinion for different scenarios

Approach to treatment			
Patient VAS	Phenotype	Treatment	Consensus
≥5	IAR or PER	Yes	Step-up
≥2 to <5	IAR	Yes	Continue
<2	IAR	Yes	Step-down
≥2 to <5	PER	Yes	Continue or Step-up
<2	PER	Yes	Step-down
≥5	IAR	No	Initiate
≥5	PER	No	Initiate
<5	IAR or PER	No	Initiate

Summary

1. RA penyakit **inflamasi di mukosa hidung**, diperantarai imunoglobulin E (IgE) akibat **pajanan allergen**
2. Keluhan RA: watery rinorea, bersin paroksismal, hidung buntu, gatal hidung dan atau mata
3. Faktor resiko yang meningkatkan RA dari pre natal, post natal dan lingkungan
4. Etiologi dari paparan allergen
5. Diagnosis yang tepat, menentukan faktor pemicu, akan mengurangi resiko komorbid pada RA

You don't realize how
much you take breathing
for granted

*until your nose
is stuffed up.*

Thank
You

